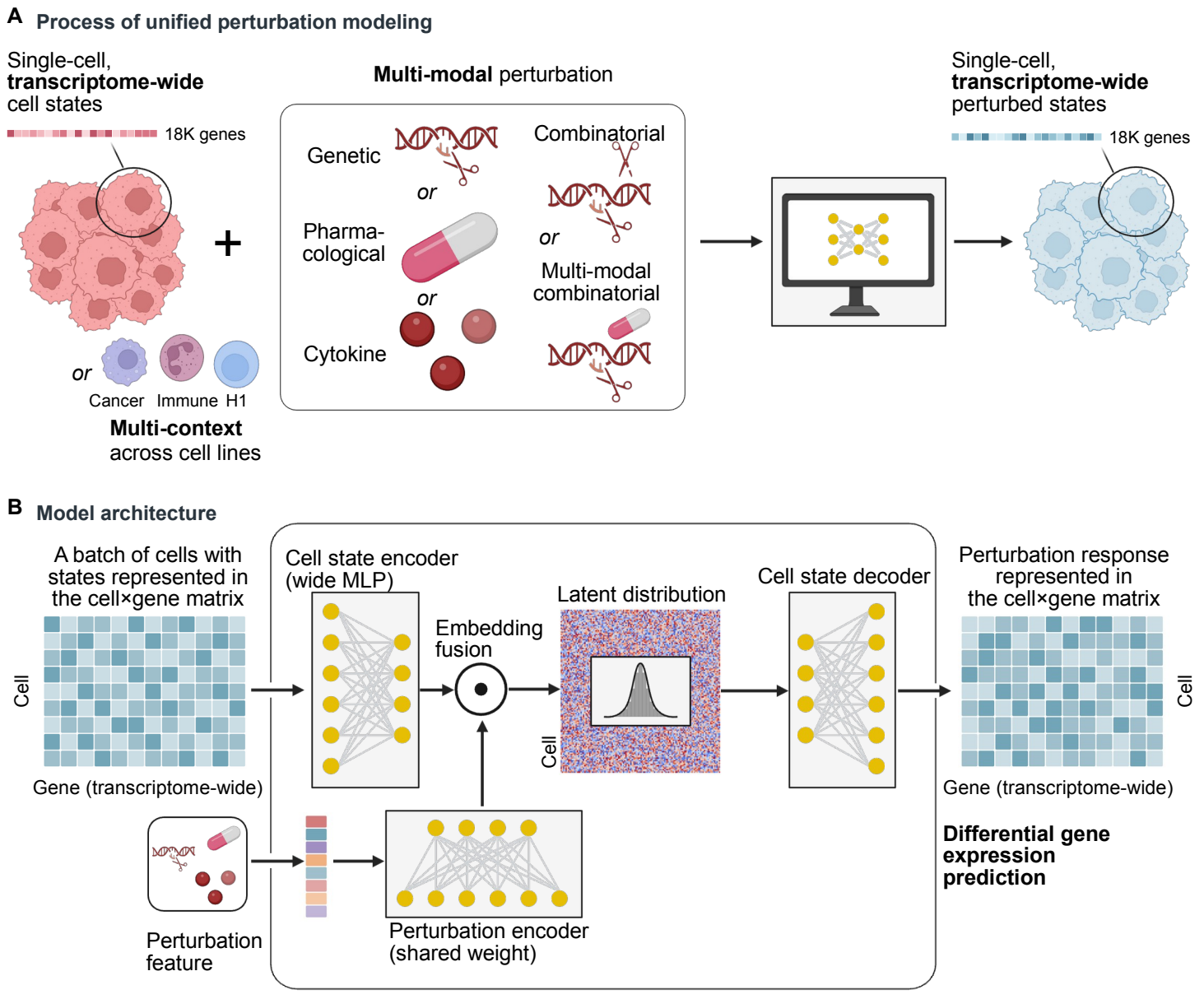

A
Process of unified perturbation modeling
Single-cell,
transcriptome-wide
cell states
18K genes
or
Cancer
Immune
H1
Multi-context
across cell lines
Multi-modal perturbation
Genetic
or
Pharma-
cological
or
Cytokine
Combinatorial
or
Multi-modal
combinatorial
Single-cell,
transcriptome-wide
perturbed states
18K genes
B
Model architecture
A batch of cells with
states represented in
the cell×gene matrix
Cell
Gene (transcriptome-wide)
Cell state encoder
(wide MLP)
Embedding
fusion
Latent distribution
Cell
Cell state decoder
Perturbation response
represented in
the cell×gene matrix
Cell
Gene (transcriptome-wide)
Differential gene
expression
prediction
Perturbation
feature
Perturbation encoder
(shared weight)